Sample Answers to Module 1: Evaluation Questions

Short-answer and scenario-based reference responses

SHORT ANSWER QUESTIONS

01 In your own words, define Machine Learning. Provide one example of how it is used in a STEM field.

> **Sample Answer:** Machine Learning is a subset of AI where systems learn patterns from data. In healthcare, ML can detect diseases like cancer by analyzing medical images.

02 Describe what an intelligent agent is. Give an example from a real-world system.

> **Sample Answer:** An intelligent agent perceives its environment through sensors and acts through actuators to achieve a goal. For example, a Roomba vacuum navigates and cleans based on its environment.

03 What is one ethical challenge that arises from using AI in hiring or admissions?

> **Sample Answer:** One challenge is algorithmic bias. If training data reflects past inequalities, AI may unfairly favor or disfavor certain groups, leading to discrimination.

04 List two differences between Machine Learning and Deep Learning.

> **Sample Answer:** ML uses algorithms that need feature selection by humans; DL uses neural networks that learn features automatically. DL requires more data and computing power than ML.

Scenario-Based Questions

Sample responses for ethics and social-impact scenarios

01 SCENARIO-BASED QUESTION

Scenario: A self-driving car must choose between hitting a pedestrian or swerving into a wall, potentially harming the driver.

Question: Discuss this from an AI ethics perspective. What principles should guide this decision, and who should be accountable?

> **Sample Answer:** Ethical frameworks like utilitarianism or the precautionary principle can guide the decision. Developers must ensure transparency and fairness in decision logic. Responsibility may fall on designers, manufacturers, and regulatory bodies.

02 SCENARIO-BASED QUESTION

Scenario: A school uses AI to automatically assign grades based on student behavior and performance data.

Question: What are some ethical concerns with this approach?

> **Sample Answer:** Concerns include lack of transparency, potential bias in grading, over-reliance on non-academic data, and students not having the ability to challenge grades.

03 SCENARIO-BASED QUESTION

Scenario: A facial recognition system in a public space is used for law enforcement.

Question: What are two potential benefits and two ethical risks of this use of AI?

> **Sample Answer:** Benefits: faster identification of suspects, public safety. Risks: invasion of privacy, misidentification—especially among minority groups.

04 SCENARIO-BASED QUESTION

Scenario: A company uses an AI chatbot to replace human customer service agents.

Question: What are the potential social and economic impacts of this decision?

> **Sample Answer:** Economically, it reduces costs, but may lead to job losses. Socially, customers may face frustration if the bot cannot understand complex requests.